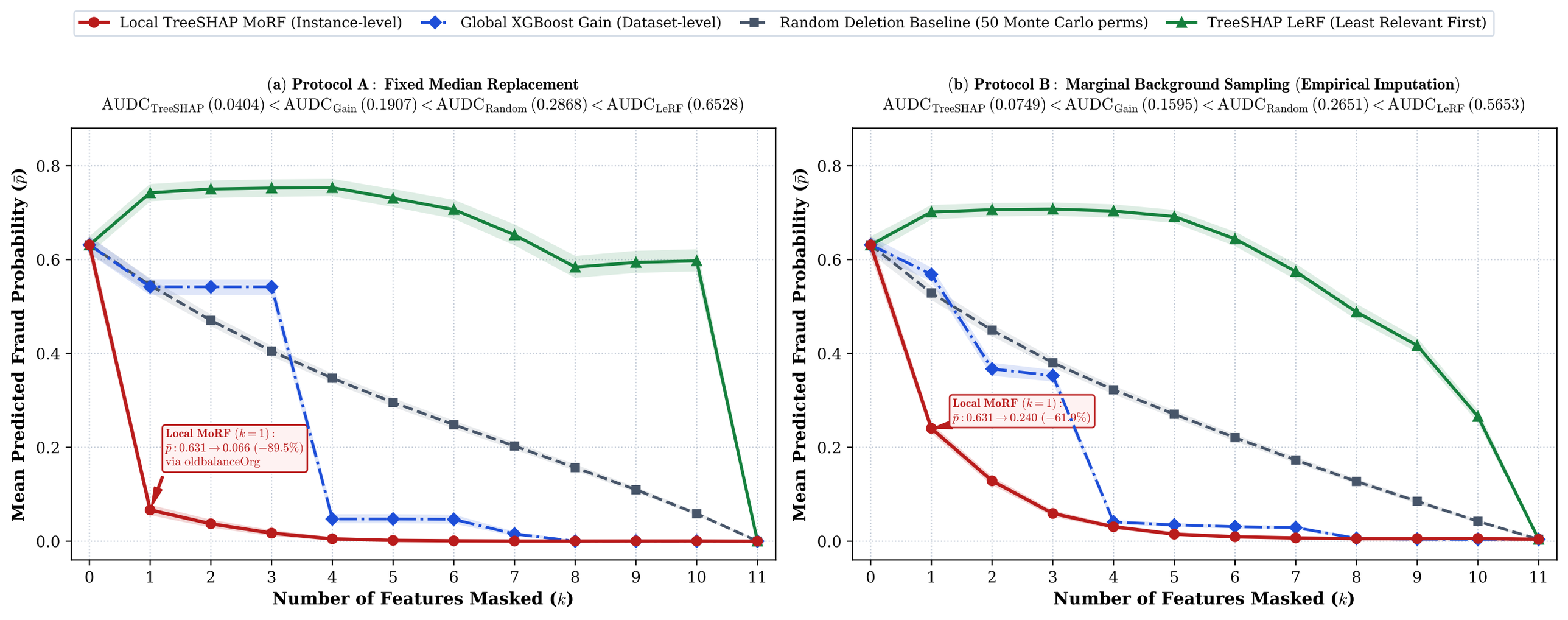

Local TreeSHAP MoRF (Instance-level)
Global XGBoost Gain (Dataset-level)
Random Deletion Baseline (50 Monte Carlo perms)
TreeSHAP LeRF (Least Relevant First)
(a) Protocol A : Fixed Median Replacement
$\mathrm{AUDC}_{\mathrm{TreeSHAP}}\,(0.0404) < \mathrm{AUDC}_{\mathrm{Gain}}\,(0.1907) < \mathrm{AUDC}_{\mathrm{Random}}\,(0.2868) < \mathrm{AUDC}_{\mathrm{LeRF}}\,(0.6528)$
(b) Protocol B : Marginal Background Sampling (Empirical Imputation)
$\mathrm{AUDC}_{\mathrm{TreeSHAP}}\,(0.0749) < \mathrm{AUDC}_{\mathrm{Gain}}\,(0.1595) < \mathrm{AUDC}_{\mathrm{Random}}\,(0.2651) < \mathrm{AUDC}_{\mathrm{LeRF}}\,(0.5653)$
Mean Predicted Fraud Probability ($\bar{p}$)
Number of Features Masked (k)
Local MoRF ($k=1$) :
$\bar{p}: 0.631 \to 0.066$ (-89.5%)
via oldbalanceOrg
Local MoRF ($k=1$) :
$\bar{p}: 0.631 \to 0.240$ (-61.9%)